2.8 Subspaces of $\mathbb{R}^n$

Definition: A subspace of $\mathbb{R}^n$ is any set H in $\mathbb{R}^n$ that has three properties:

a. The zero vector is in H.

b. For each u and v in H, the sum $u+v$ is in H.

c. For each u in H and each scalar c, the vector cu is in H.

In words, a subspace is closed under addition and scalar multiplication.

Ex: If v_1 and v_2 are in $\mathbb{R}^n$ and $H = \text{Span}\{v_1, v_2\}$, then H is a subspace of $\mathbb{R}^n$.

Proof/Solution:

a. Zero vector is in H because $\underbrace{0v_1 + 0v_2}_{=0}$ is a linear combination of v_1 and v_2.

b. Take two vector in H, say

$u = s_1v_1 + s_2v_2$, $v = t_1v_1 + t_2v_2$ clearly u and v are in $\text{Span}\{v_1, v_2\}$

Then

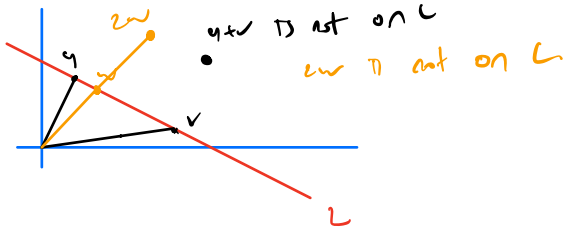
$u+v = (s_1+t_1)v_1 + (s_2+t_2)v_2$, clearly $u+v$ is in $\text{Span}\{v_1, v_2\}$

Hence, $u+v$ is in H

c. For any scalar c, the vector cu is in H because

$cu = c(s_1v_1 + s_2v_2) = (cs_1)v_1 + (cs_2)v_2$.

- If v_1 is not zero and v_2 is a multiple of v_1, then v_1 and v_2 simply span a line through the origin. So, a line through the origin is another example of a subspace.
- A line L not through the origin is not a subspace because it does not contain the origin.



Ex: For $v_1, \dots, v_p$ in $\mathbb{R}^n$, the set of all linear combinations of $v_1, \dots, v_p$ is a subspace of $\mathbb{R}^n$. We shall now refer to $\text{Span}\{v_1, \dots, v_p\}$ as the subspace spanned (or generated) by $v_1, \dots, v_p$.

- Note that $\mathbb{R}^n$ is a subspace of itself.

Column space and Null space of a matrix

Def: The column space of a matrix A is the set $\boxed{\text{Col}\,A}$ of all linear combinations of the columns of A.

If $A = [a_1 \ \dots \ a_n]$ with the columns in $\mathbb{R}^m$, then $\text{Col}\,A$ is the same as $\text{Span}\{a_1, \dots, a_n\}$. Column space of an $m \times n$ matrix is a subspace of $\mathbb{R}^m$.

Ex: Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$ $b = \begin{bmatrix} 3 \\ 3 \\ -4 \end{bmatrix}$ Is b in the column space of A?

Solution: The vector b is a linear combination of the columns of A iff b can be written as Ax for some x. I.e. iff the equation $Ax = b$ has a solution.

Then

$$\begin{bmatrix} 1 & -3 & -4 & 3 \\ -4 & 6 & -2 & 3 \\ -3 & 7 & 6 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & -2 & -6 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 & -4 & 3 \\ 0 & -6 & -18 & 15 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

We can conclude that $Ax = b$ is consistent and b is in $\text{Col}A$.

$0x_3 = 0 \rightarrow x_3 = 0$

- consistent
- infinitely many solutions.

So, like a system of linear equations is written in the form $Ax = b$, the column space of A is the set of all b for which the system has a solution.

Def: The null space of a matrix A is the set $\text{Nul}A$ of all solutions of the homogeneous equation $Ax = 0$.

Theorem: The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently the set of all solutions of a system $Ax = 0$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

To test whether a given vector v is in the NulA, just compute Av to see whether Av is the zero vector.

Basis for a subspace

Because a subspace typically contains an infinite number of vectors, some problems involving a subspace are handled best by working with a small finite set of vectors that span the subspace. The smaller the set, the better.

Def: A basis for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H.

Ex: The columns of an invertible $n \times n$ matrix form a basis for all of $\mathbb{R}^n$ because they are linearly independent and span $\mathbb{R}^n$, by the invertible matrix theorem. One such matrix is the $n \times n$ identity matrix. Its columns are denoted by $e_1, \dots, e_n$:

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

The set $\{e_1, \dots, e_n\}$ is called the standard basis for $\mathbb{R}^n$.

Ex. Find a basis for the null space of the matrix:

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}_{3\times 5}$$

By the above Theorem the Nul A is a subspace of $\mathbb{R}^5$.

Sol: First write a solution of $Ax = 0$ in parametric vector form:

$$\left[\begin{array}{ccccc|c} A & & & & & 0 \end{array}\right] \sim \left[\begin{array}{ccccc|c} \boxed{1} & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & \boxed{1} & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

x_1 x_2 x_3 x_4 x_5

basic free basic free free

$$\Rightarrow \begin{aligned} x_1 - 2x_2 \quad - x_4 + 3x_5 &= 0 \\ x_3 + 2x_4 - 2x_5 &= 0 \\ 0 &= 0 \end{aligned}$$

So, the general solution is $x_1 = 2x_2 + x_4 - 3x_5$

$x_3 = -2x_4 + 2x_5$

x_2, x_4, x_5 = free

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

u v w

$$= x_2 u + x_4 v + x_5 w. \quad (1)$$

let $x_2 = x_4 = x_5 = 1$ → Then $x = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$

check $Ax = 0$

$$A \cdot x = 0$$

$$\begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Basis for $\mathrm{Nul}\,A$

$\{u, v, w\}$ generates $\mathrm{Nul}\,A$. They are linearly independent because Eq (1) shows that $0 = x_2 u + x_4 v + x_5 w$ only if the weights x_2, x_4 and x_5 are all zero.

Ex: Find a basis for the column space of the matrix:

$$B = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

b_1 b_2 b_3 b_4 b_5

e_1 e_2 e_3

Sol. See that $b_3 = -3b_1 + 2b_2$ So, $\{b_1, b_2, b_5\}$ spans $\mathrm{Col}\,B$.

$b_4 = 5b_1 - b_2$

Also b_1, b_2, b_5 are linearly independent because they are columns from an identity matrix.

So, the pivot columns of B form a basis for $\mathrm{Col}\,B$.

Ex: It can be verified that the matrix

$$A = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 & a_5 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 1 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix}$$

is row equivalent to the matrix B in the previous example. Find a basis for $\operatorname{Col} A$.

Sol Remember $b_3 = -3b_1 + 2b_2$, $b_4 = 5b_1 - b_2$. Since row operations do not affect linear dependence relations among the columns of the matrix, we should have $a_3 = -3a_1 + 2a_2$, $a_4 = 5a_1 - a_2$. Check that this is true!

Since $\{b_1, b_2, b_5\}$ is linearly independent, also linearly independent. Hence $\{a_1, a_2, a_5\}$ is a basis for $\operatorname{Col} A$.

Theorem: The pivot columns of a matrix A form a basis for the column space of A.